

Highlights

A boundary-integrated closure for renormalised meshless operators on elliptic boundary value problems

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- A unified boundary closure integrates Neumann and Dirichlet or Robin conditions into renormalised meshless operators on one code path.
- The free surface is detected from the point cloud and enforced without a tuning parameter, the enforcement strength cancelling to the total kernel weight.
- Non-orthogonal corners are treated exactly by a general Gram projector applied in physical space before renormalisation.
- An algebraic-ghost closure restores Neumann accuracy without extra unknowns, and with the normalisation it approaches second-order convergence.
- In a Helmholtz–Hodge decomposition the operators are consistent, the residual divergence being their non-adjointness defect, but vanishing with refinement.
- Iterating the decomposition is preconditioned Richardson on the non-adjoint divergence–gradient pairing; a minimal-residual step makes it non-expansive and stable where the fixed-step iteration diverges.

A boundary-integrated closure for renormalised meshless operators on elliptic boundary value problems

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Abstract

A boundary-condition closure for renormalised, least-squares corrected meshless Laplacian operators is presented. In contrast to treatments that pad the domain with auxiliary particles, embed it in a level set, or enforce the condition through a weak form, the closure is algebraic and strong-form: the Neumann and free-surface conditions are folded directly into the discrete collocation row on a single code path, with the boundary detected from the point cloud and without tuning parameters. The closure is organised by the condition type. A Neumann flux enters the right-hand side, and a Dirichlet or Robin value enters the diagonal. The Neumann flux is imposed by a tangential projection performed in physical space before renormalisation. A general Gram projector treats non-orthogonal corners exactly, and a complementary algebraic-ghost closure completes the truncated boundary stencil by reflecting the neighbours and the node itself across the local reflection group, without extra unknowns. The free surface is detected from the support-deficiency geometry, and its enforcement strength cancels to the total kernel weight, which removes any penalty or threshold parameter. The introduced method, named FIRM (Flux-Integrated Renormalised Meshless),

is verified against analytic ground truths and on irregular-domain Poisson benchmarks, for regular and highly irregular point clouds in two dimensions, against a second-order generalised finite difference baseline. The operators are linear-exact on any cloud. The value closure supraconverges, and the algebraic-ghost closure reduces the Neumann-limited error by about an order of magnitude and, with the sum normalisation, approaches second-order convergence at a straight Neumann boundary. Used in a Helmholtz–Hodge decomposition, the renormalised operators are shown to be consistent for a single decomposition, the residual divergence equalling the operators’ non-adjointness defect and vanishing under refinement; the fixed-step iteration of the decomposition is preconditioned Richardson on the non-adjoint pairing and can diverge, whereas a minimal-residual step renders it non-expansive and stable on disordered or time-evolving clouds.

Keywords: meshless method, boundary conditions, renormalised Laplacian, free surface, Neumann, Poisson equation

1. Introduction

The numerical solution of elliptic boundary value problems on irregular domains is a recurring requirement across computational physics. It is especially demanding for meshless discretisations in which the computational nodes move and the domain boundary is not represented by a conforming mesh, as in the pressure projection of incompressible free-surface flow and, more generally, in diffusion, electrostatics, and porous-media seepage on irregular or evolving domains. In a companion study, a class of renormalised meshless Laplacian operators was introduced for the Poisson and diffusion

equations [1]. These operators are corrected by a local least-squares moment matrix and remain consistent on highly irregular point arrangements, where the classical smoothed-particle-hydrodynamics (SPH) Laplacian fails to converge [2, 3, 4]. Their least-squares correction belongs to the kernel-gradient renormalisation lineage developed to restore particle consistency [5, 6, 7]. That study, however, treated the interior operator and reported Dirichlet and Robin results on prescribed domains only; it left open how a boundary condition is integrated into the renormalised operator when the boundary is detected from the cloud rather than prescribed, and how Neumann and free-surface boundaries are handled on a single code path. The present work closes this boundary gap, and is orthogonal to the recent extension of the same interior operator to full second-order consistency on disordered clouds by Asai et al. [8].

Handling boundaries on meshless or otherwise irregular domains has been approached through several distinct methodological families, which differ less in accuracy than in what each requires of the geometry. A first family represents the boundary geometrically and embeds it in a background grid. In the level-set tradition, Gibou and co-workers built second-order-accurate symmetric discretisations of the Poisson equation on irregular domains through a ghost-value treatment [9, 10, 11, 12], extended to Robin conditions on implicit interfaces [13] and to Voronoi interface constructions [14]; these methods are accurate and well analysed, but they require an adaptive Cartesian grid and a level-set description of the geometry. A second family represents the boundary with auxiliary material. In SPH and ISPH, solid walls and free surfaces are carried by dummy or mirror particles, repulsive forces, or

generalised wall conditions [15, 16, 17], by semi-analytical boundary integrals [18, 19, 20], or by the corrected Laplacians of the moving-particle semi-implicit family [21, 22, 23, 24, 25]; these treatments are effective but typically introduce auxiliary particles and a tuning length or threshold, the free surface in particular being identified by a thresholded density or kernel-sum criterion. A third family appends the boundary condition as additional algebraic rows: in the generalised finite difference (GFDM) and finite-pointset (FPM) methods a full Taylor basis is fitted by weighted least squares and the conditions enter as constraint rows [26, 27, 28, 29], while radial-basis-function finite differences recover boundary accuracy by adding ghost nodes outside the domain [30, 31, 32]. Across these families the boundary is paid for with stored degrees of freedom, a background grid, or a tuning parameter.

A separate line of work injects the boundary condition into the discrete equations *algebraically*, rather than padding the domain with stored geometry, and three strands may be distinguished. The first embeds the condition in the trial space *a priori*, so that every admissible field satisfies it by construction: the R-function method builds boundary-conforming solution structures from implicit geometry [33, 34], weighted extended B-splines weight a tensor-product basis by a boundary distance function [35], and the same construction underlies the exact imposition of boundary conditions in neural and physics-informed solvers through distance functions [36, 37]. The second injects the condition into the variational form rather than the basis: in element-free Galerkin and related meshfree methods the moving-least-squares shape functions lack the Kronecker-delta property and so cannot pin nodal values directly [38, 39], and essential conditions are instead enforced weakly

through Nitsche’s method [40] or a penalty [41], with strong-type variants developed more recently [42]. The third does not prescribe a condition at all but corrects the truncated kernel summation near the boundary so that the operator stays consistent there, through Shepard normalisation [43], corrective and reproducing-kernel formulations [44, 45], and the boundary truncation-error analysis of Quinlan et al. [46]; this strand is the closest ancestor of the renormalised operators used here, though it restores consistency without prescribing the boundary value or flux. The present closure is algebraic in this sense but differs from all three: it embeds the boundary condition neither in the trial space nor in a variational form requiring background integration, but directly into a strong-form collocation row, organised by condition type, with a Neumann flux entering the right-hand side and a Dirichlet or Robin value the diagonal of the same row. The boundary is detected from the point cloud, and the closure carries no tuning parameter.

The contribution of this work is a boundary closure for the renormalised operators of [1] with three properties not jointly provided by the methods reviewed above. It is *unified*: the interior, the Neumann boundary, the free surface and the contact lines are assembled by one kernel on a single code path, distinguished only by the boundary-condition type, a Neumann flux entering the right-hand side and a Dirichlet or Robin value the diagonal of the same collocation row. It is *geometry-detected*: the free surface is identified from the support-deficiency geometry of the cloud, without surface particles and without a level set. It is *parameter-free*: the free-surface enforcement strength cancels to the total kernel weight, a geometric quantity, and a parameter-free activation leaves the interior nodes untouched, so the

closure introduces no penalty, threshold, or tuning length. A complementary algebraic-ghost closure removes the accuracy bottleneck of the flux-only Neumann treatment by completing the truncated boundary stencil with reflected samples that are formed on the fly and never stored, and so add no unknowns. As a smaller supporting result, the single-sum interior operator adopted here, which we call the *smooth-naive* operator, bridges the naive and sum versions of [1] through a trace-based normalisation and retains the corrected summation of the latter in a single neighbour loop. The scope is restricted to the linear elliptic Poisson problem; applications and extensions are discussed in the conclusions.

The paper is organised as follows. Section 2 recalls the renormalised operators. Section 3 derives the boundary closure, comprising the boundary-condition-by-type principle, the Neumann projection and its corner treatment, the algebraic-ghost closure, the free-surface detection, and a short consistency remark. Section 4 reports the pointwise operator-approximation results, including boundary and corner rows. Section 5 reports the boundary-value-problem benchmarks and the comparison with the GFDM baseline. Section 6 applies the closure to the Helmholtz–Hodge decomposition and examines the operators’ non-adjointness. Section 7.1 discusses computational aspects, and Section 8 concludes.

2. Meshless operators

Three meshless second-derivative operators are used in this work and are set out in this section in a common notation: the renormalised differencing operator of [1], abbreviated LDD after the Lagrangian differencing dynamics

of [47], on which the boundary closure of Section 3 is built; a generalised finite difference (GFDM) operator used as a second-order baseline; and the second-order renormalised operator of [8], which we re-express in a kernel-gradient-free form consistent with the present framework. For a point cloud Ω with nodal spacing Δ , each node i interacts with neighbours j within a compact support radius $h = \kappa\Delta$ through a kernel $W_{ij} = W(|\mathbf{x}_{ij}|, h)$, where $\mathbf{x}_{ij} = \mathbf{x}_j - \mathbf{x}_i$ and $f_{ij} = f_j - f_i$. The kernel used throughout is the poly6 function

$$W(r, h) = \max(0, 1 - q^2)^3, \quad q = r/h, \quad (1)$$

with $\kappa = 2.5$ by default.

2.1. Lagrangian Differencing Dynamics (LDD)

The renormalised operators of [1] consume the kernel weights and renormalise them, and are therefore kernel-agnostic in the sense made precise below. From the kernel are formed the support-deficiency vector $\mathbf{o}_i$, the total weight S_i , the second-moment renormalisation tensor $\mathbf{M}_i$ and its inverse $\mathbf{B}_i$,

$$\mathbf{o}_i = \sum_j W_{ij} \mathbf{x}_{ij}, \quad (2)$$

$$S_i = \sum_j W_{ij}, \quad (3)$$

$$\mathbf{M}_i = \sum_j W_{ij} \mathbf{x}_{ij} \otimes \mathbf{x}_{ij}, \quad (4)$$

$$\mathbf{B}_i = \mathbf{M}_i^{-1}. \quad (5)$$

The vector $\mathbf{o}_i$ vanishes for symmetric support and points inward near a boundary, and $\mathbf{B}_i$ is symmetric positive-definite whenever $\mathbf{M}_i$ has full rank,

which requires at least $d + 1$ non-degenerate neighbours. The renormalised gradient is

$$\langle \nabla f \rangle_i = \mathbf{B}_i \sum_j W_{ij} f_{ij} \mathbf{x}_{ij}, \quad (6)$$

which is exact for linear fields on any cloud. For $f = \mathbf{a} \cdot \mathbf{x} + b$ the difference is $f_{ij} = \mathbf{a} \cdot \mathbf{x}_{ij}$, and by the definition of $\mathbf{M}_i$ in Eq. (4),

$$\begin{aligned} \langle \nabla f \rangle_i &= \mathbf{B}_i \sum_j W_{ij} (\mathbf{a} \cdot \mathbf{x}_{ij}) \mathbf{x}_{ij} \\ &= \mathbf{B}_i \left(\sum_j W_{ij} \mathbf{x}_{ij} \otimes \mathbf{x}_{ij} \right) \mathbf{a} = \mathbf{B}_i \mathbf{M}_i \mathbf{a} = \mathbf{a}, \end{aligned} \quad (7)$$

the renormalisation by $\mathbf{B}_i = \mathbf{M}_i^{-1}$ being precisely what cancels the unbalanced moments of a truncated support.

The Laplacian is again obtained from a Taylor expansion in which the first-order term is removed by subtracting the reconstructed gradient. Two members of the renormalised family derived in [1] provide the context. The *naive version* weights each neighbour by the reciprocal squared distance,

$$\langle \nabla^2 f \rangle_i^{\text{naive}} = 2d \sum_j \frac{W_{ij}}{|\mathbf{x}_{ij}|^2} (f_{ij} - \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i), \quad (8)$$

which is accurate on regular clouds but is sensitive to close neighbours, since a single near pair makes $1/|\mathbf{x}_{ij}|^2$ large. The *sum version* removes that sensitivity by replacing the per-neighbour weight with a single self-normalising ratio,

$$\langle \nabla^2 f \rangle_i^{\text{sum}} = 2d \frac{\sum_j W_{ij} w_{ij} f_{ij}}{\sum_j W_{ij} w_{ij} |\mathbf{x}_{ij}|^2}, \quad w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i, \quad (9)$$

whose denominator is the operator's own action on $\frac{1}{2}|\mathbf{x}|^2$ and which is therefore exact for the isotropic quadratic by construction.

The operator introduced here, which we name the *smooth-naive* operator, descends from the naive form, Eq. (8), by smoothing its normalisation. The per-neighbour factor $2d/|\mathbf{x}_{ij}|^2$ is replaced by a single tensor quantity built from the already-computed renormalisation tensor,

$$N_i = \frac{2}{d} \text{tr } \mathbf{B}_i. \quad (10)$$

It was found that $\text{tr } \mathbf{B}_i$ represents the normalisation more smoothly than the per-neighbour reciprocal, being insensitive to individual close pairs. The factor $2/d$ follows from the spherical-mean identity $\nabla^2 f = \lim_{r \rightarrow 0} \frac{2d}{r^2} (\bar{f}_r - f)$ and gives the correct second-derivative magnitude in any dimension, as confirmed by a three-dimensional constant-Laplacian test in Section 4. With this normalisation the corrected second difference reads

$$\langle \nabla^2 f \rangle_i = N_i \sum_j W_{ij} (f_{ij} - \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i). \quad (11)$$

Because $\langle \nabla f \rangle_i$ does not depend on j , the offset $\sum_j W_{ij} \mathbf{x}_{ij} = \mathbf{o}_i$ is factored out of the second term, giving $\sum_j W_{ij} \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i = \mathbf{o}_i \cdot \langle \nabla f \rangle_i$. Substituting Eq. (6) and using the symmetry of $\mathbf{B}_i$,

$$\begin{aligned} \mathbf{o}_i \cdot \langle \nabla f \rangle_i &= \mathbf{o}_i \cdot \mathbf{B}_i \sum_j W_{ij} f_{ij} \mathbf{x}_{ij} \\ &= \sum_j W_{ij} f_{ij} (\mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i), \end{aligned} \quad (12)$$

so the double sum collapses to the single-sum renormalised Laplacian

$$\langle \nabla^2 f \rangle_i = N_i \sum_j W_{ij} f_{ij} (1 - \mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i) \equiv N_i \sum_j W_{ij} f_{ij} w_{ij}. \quad (13)$$

Equation (13) connects the two operators of [1]. Its summand $W_{ij} w_{ij} f_{ij}$ is the numerator of the sum version, Eq. (9), while its normalisation N_i

is the smoothed descendant of the naive weighting, Eq. (8). The smooth-naive operator therefore retains the corrected, offset-shifted summation of the sum version, and evaluates it in a single neighbour loop without the second summation. The three normalisations moreover coincide on isotropic support, where $w_{ij} \rightarrow 1$ and $\mathbf{M}_i = \frac{1}{d}(\sum_j W_{ij}|\mathbf{x}_{ij}|^2)\mathbf{I}$, giving

$$N_i^{\text{den}} \equiv \frac{2d}{\sum_j W_{ij} w_{ij} |\mathbf{x}_{ij}|^2} \xrightarrow{\text{isotropic}} \frac{2d}{\sum_j W_{ij} |\mathbf{x}_{ij}|^2} = \frac{2}{d} \text{tr } \mathbf{B}_i = N_i, \quad (14)$$

the naive, sum and smooth-naive operators all reducing to the same expression on a regular lattice and differing only under anisotropy. The self-normalising denominator N_i^{den} of Eq. (14) is the normalisation of the *sum* operator of [1]; applied to the same summand it is retained here as the more accurate variant. Being exact for $|\mathbf{x}|^2$, the sum operator is between 8 and 15% more accurate than the smooth-naive operator under disorder, with the edge widening as the chaos level grows; the sum-to-smooth-naive error ratio falls from 1.00 on a lattice to 0.73 at chaos 80%, as shown in Section 5, Fig. 12. The smooth-naive operator, which avoids the denominator sum, is the default, and the sum operator is offered as a more accurate refinement.

The correction weight renders Eq. (13) linear-exact on any cloud, including the truncated rows that occur at a boundary. For $f = \mathbf{a} \cdot \mathbf{x} + b$ the field difference is again $f_{ij} = \mathbf{a} \cdot \mathbf{x}_{ij}$, and

$$\begin{aligned} \langle \nabla^2 f \rangle_i &= N_i \left[\sum_j W_{ij} (\mathbf{a} \cdot \mathbf{x}_{ij}) - \sum_j W_{ij} (\mathbf{a} \cdot \mathbf{x}_{ij}) (\mathbf{x}_{ij} \cdot \mathbf{B}_i \mathbf{o}_i) \right] \\ &= N_i [\mathbf{a} \cdot \mathbf{o}_i - \mathbf{a} \cdot \mathbf{M}_i \mathbf{B}_i \mathbf{o}_i] = 0, \end{aligned} \quad (15)$$

since $\mathbf{M}_i \mathbf{B}_i = \mathbf{I}$, and the two terms cancel on any support, symmetric or truncated. Readers interested in the rigorous derivation and the convergence theory of these operators are referred to [1, 48, 49].

It should be noted that Eq. (13) is linear-consistent but not, in general, second-order consistent on a disordered cloud. On a regular lattice its pointwise truncation error is $O(\Delta^2)$, whereas under fixed-relative jitter it is $O(1)$ and does not vanish as $\Delta \rightarrow 0$. The Poisson solution nonetheless converges, a behaviour known as supraconvergence, because the $O(1)$ pointwise error is the discrete divergence of a consistent flux and is damped by the global solve. This behaviour was established in [1]. It frames the accuracy results of the present study and is revisited in Section 3.6.

2.2. Generalised finite differences (GFDM)

As a second-order reference we use a generalised finite difference operator of the finite-pointset type [26, 27, 28]. At each node the full quadratic Taylor basis on the neighbour offsets,

$$\mathbf{p}_{ij} = \left[x_{ij}, y_{ij}, \frac{1}{2}x_{ij}^2, x_{ij}y_{ij}, \frac{1}{2}y_{ij}^2 \right]^\top, \quad (16)$$

is fitted to the field differences f_{ij} by weighted least squares with the same kernel weights W_{ij} , and the Laplacian is read off as the sum of the two pure second-derivative coefficients,

$$\langle \nabla^2 f \rangle_i^{\text{GFDM}} = \mathbf{e}^\top (\mathbf{P}^\top \mathbf{W} \mathbf{P})^{-1} \mathbf{P}^\top \mathbf{W} \mathbf{f}, \quad \mathbf{e} = [0, 0, 1, 0, 1]^\top, \quad (17)$$

with $\mathbf{P}$ the matrix of basis rows $\mathbf{p}_{ij}^\top$, $\mathbf{W} = \text{diag}(W_{ij})$ and $\mathbf{f} = (f_{ij})_j$. The operator is exact for quadratics and hence second-order consistent in the interior, on regular and disordered clouds alike, at the cost of a 5×5 local solve in two dimensions and a requirement of at least five well-spread neighbours; on a regular grid it reduces to the standard five-point stencil. The Neumann

flux is imposed as an exact equality constraint through a local Karush–Kuhn–Tucker saddle system, in the manner of the finite-pointset constraint row [27, 28], and a soft penalty-flux variant, shown not to converge in Section 5.2, is referred to as the penalty form.

2.3. Second-order operators in LDD form

The cross-term free LDD operator in Section 2.1 are linear-consistent. In a systematic taxonomy of renormalised second-derivative models, Asai et al. [8] identify their *Block Diagonal* operator, equivalent there to the corrected Laplacian of Schwaiger [50] and to the present family, with their *Naive* variant coinciding with the Taylor-series form of Lian et al. [51]. Full second-order pointwise consistency under disorder is recovered only by their *Full Inverse* operator, which reinstates the cross-derivative coupling that the block-diagonal reduction drops, in common with the consistent mesh-less second-derivative constructions of Español and Revenga [52], the least-squares moving-particle method of Tamai and Koshizuka [53], and the GFDM operator of Section 2.2. The terminology is not uniform across the literature, i.e. what [8] call Block Diagonal, is the variant termed *Full Inverse* in the LDD line of work [1, 47].

This operator attains second order by fitting the full set of second derivatives, including the cross term, rather than the trace alone. It is written in terms of a corrected kernel gradient $\tilde{\nabla}W_{ij}$ and the weighting $\mathbf{r}_{ij} \cdot \tilde{\nabla}W_{ij}/|\mathbf{r}_{ij}|^4$. The present framework, however, never differentiates the kernel of the gradient (6) and the Laplacian (13), that use only the scalar weight W_{ij} and the renormalisation $\mathbf{B}_i$. We therefore apply the idea to an equivalent, kernel-gradient-free form of the operator, expressed in the same renormalise-by-

$\mathbf{B}_i$ idiom as a single weighted least-squares fit. Collect the second derivatives in $\mathbf{u}_i = [\partial_{xx}^2 f, \partial_{yy}^2 f, 2\partial_{xy}^2 f]^\top$ and the quadratic offsets in the basis $\mathbf{s}_{ij} = [x_{ij}^2, y_{ij}^2, x_{ij}y_{ij}]^\top$. The renormalised gradient (6) is the stencil $\langle \nabla f \rangle_i = \sum_j \mathbf{c}_{ij} f_{ij}$ with $\mathbf{c}_{ij} = W_{ij} \mathbf{B}_i \mathbf{x}_{ij}$. Removing it from the second difference leaves the second-order remainder

$$g_{ij} \equiv f_{ij} - \mathbf{x}_{ij} \cdot \langle \nabla f \rangle_i = \frac{1}{2} \hat{\mathbf{s}}_{ij} \cdot \mathbf{u}_i, \quad (18)$$

exact for quadratics, in which the basis must be *corrected* for the second-derivative content that the renormalised gradient itself carries,

$$\hat{\mathbf{s}}_{ij} = \mathbf{s}_{ij} - \mathbf{C}_i \mathbf{x}_{ij}, \quad \mathbf{C}_i = \sum_j W_{ij} \mathbf{s}_{ij} (\mathbf{B}_i \mathbf{x}_{ij})^\top. \quad (19)$$

The term $\mathbf{C}_i \mathbf{x}_{ij}$ is the kernel-gradient-free counterpart of the $R_{(2)}$ correction of [8]; omitting it collapses the fit to the linear-consistent class and the second order is lost under disorder. The coefficients follow from the weighted normal equations

$$\mathbf{M}_i^{\text{FI}} \mathbf{u}_i = 2 \sum_j W_{ij} \hat{\mathbf{s}}_{ij} g_{ij}, \quad \mathbf{M}_i^{\text{FI}} = \sum_j W_{ij} \hat{\mathbf{s}}_{ij} \otimes \hat{\mathbf{s}}_{ij}, \quad (20)$$

a 3×3 system in two dimensions, and the Laplacian is the trace $\langle \nabla^2 f \rangle_i^{\text{FI}} = u_{i,1} + u_{i,2}$. Dropping the cross column of $\mathbf{s}_{ij}$ recovers the block-diagonal, linear-consistent operator, and hence Eq. (13), which is the equivalence noted above. The construction uses only W_{ij} , $\mathbf{B}_i$ and the offsets, with no kernel derivative; in practice the offsets are scaled by their root-mean-square length to condition $\mathbf{M}_i^{\text{FI}}$. This form was verified against the analytic moment integrals of [8], recovering their uniform-cloud values to the quadrature error, and against pointwise second-order convergence on disordered clouds.

The present work deliberately retains the original linear-consistent LDD operator and shows that, once the boundary is closed correctly, it is the boundary closure and not the interior order that governs the realised accuracy of the solution. The interior operators from this section and the boundary closure to be introduced are complementary and could be combined, as summarised in Table 2.

3. Boundary closure

3.1. Boundary conditions by type

The organising principle of the closure is that the type of the boundary condition, not the physical quantity it constrains, determines the algebra of its integration into Eq. (13). A Neumann condition prescribes a flux $\nabla p \cdot \mathbf{n} = g$. Since a flux is a known datum independent of the unknown nodal value, it is moved to the right-hand side. A Dirichlet or Robin condition prescribes a value, or a value-dependent flux. Since the constraint involves the unknown nodal value, it contributes to the diagonal self-interaction term as well as to the right-hand side. Two independent classifications are at work, and the closure keeps them separate. The *enforcement algebra* just described, flux to the right-hand side and value to the diagonal, is one axis; the boundary *representation*, prescribed or detected from the cloud, is the other. The two closures derived below are named for the algebra, a *flux closure* and a *value closure*, while the wall and the free surface are physical instances of them, the free surface additionally being detected rather than prescribed. The generalised Robin condition $\nabla p \cdot \mathbf{n} + \alpha p = \gamma$ unifies the algebra: the flux closure is the limit $\alpha \rightarrow 0$ and the value closure the limit

$\alpha \rightarrow \infty$, a finite α splitting the datum between the two (Section 5.5). The pairing of the wall with a flux and the free surface with a value is therefore specific to the pressure problem and reverses for the velocity field, where the no-slip wall is a value closure and the stress-free surface a flux closure (Section 8). The unified assembled row is

$$\sum_j W_{ij} w_{ij} p_{ij} - \sigma_i S_i p_i = b_i, \quad (21)$$

with $w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{V}_i$ and a correction vector $\mathbf{V}_i$ that absorbs the Neumann and free-surface geometry, derived below. In difference form the sparse matrix and the right-hand side are assembled as

$$A_{ij} = W_{ij} w_{ij} \quad (j \neq i), \quad (22)$$

$$A_{ii} = - \sum_j W_{ij} w_{ij} - \sigma_i S_i, \quad (23)$$

$$b_i = \frac{\nabla^2 p_i}{N_i} + b_{N,i} - \sigma_i S_i p_{\text{target}}. \quad (24)$$

The source enters normalised by N_i . The Neumann flux $b_{N,i}$ is already normalised and is not divided by N_i . The free-surface term contributes $-\sigma_i S_i$ to the diagonal, Eq. (23), and $-\sigma_i S_i p_{\text{target}}$ to the right-hand side, Eq. (24).

3.2. Flux closure (Neumann) by tangential projection

Two flux closures are presented for the Neumann boundary, of which this section presents the tangential projection scheme. It is linear-exact but imposes only the flux, and the resulting one-sided normal stencil limits its accuracy, which motivates the algebraic-ghost closure of Section 3.3.

Additional nodes are not generated on the Neumann boundary. The modification activates on any node whose support is truncated by the boundary,

within a distance of order h . Such a node interacts with K nearby boundary segments, each with an outward unit normal $\mathbf{n}_k$, a foot point $\mathbf{x}_{f,k}$, and a prescribed flux g_k . A tangential projector $\mathbf{P}_{\text{tan}}$ removes the boundary-normal directions from the reconstruction, and the unresolved normal contribution is supplied by the known flux. Two projectors are introduced as follows.

The *averaged-normal* projector (referred to as AN) builds a single effective normal from the boundary normals and projects it out,

$$\mathbf{n}_{\text{eff}} = \frac{\sum_k \beta_k \mathbf{n}_k}{\left| \sum_k \beta_k \mathbf{n}_k \right|}, \quad (25)$$

$$\mathbf{P}_{\text{tan}} = \mathbf{I} - \mathbf{n}_{\text{eff}} \otimes \mathbf{n}_{\text{eff}}, \quad (26)$$

where the proximity weights decay with the distance to the boundary $\delta_{w,k}$,

$$\beta_k = \frac{\max(0, 1 - \delta_{w,k}/h_w)^2}{\sum_m \max(0, 1 - \delta_{w,m}/h_w)^2}. \quad (27)$$

The AN projector is exact for a flat boundary and for smooth curved boundaries, but it introduces an $O(\Delta)$ error at a non-orthogonal corner, where a single averaged normal cannot span the full normal subspace. The *general Gram projector* (referred to as GGP) instead removes the entire subspace spanned by the boundary normals. Collecting them as the columns of $\mathbf{N} = [\mathbf{n}_1 \mid \cdots \mid \mathbf{n}_K]$, the Gram matrix and the projector are

$$\mathbf{G} = \mathbf{N}^\top \mathbf{N}, \quad (28)$$

$$\mathbf{P}_{\text{tan}} = \mathbf{I} - \mathbf{N} \mathbf{G}^{-1} \mathbf{N}^\top. \quad (29)$$

The projector is symmetric, idempotent and annihilates every boundary normal, $\mathbf{P}_{\text{tan}} \mathbf{n}_k = \mathbf{0}$, and is therefore exact for arbitrary corner geometry. Near-parallel boundaries are regularised by $\mathbf{G} \rightarrow \mathbf{G} + \epsilon \mathbf{I}$ with $\epsilon \sim 10^{-2}$. For $K = 1$

and for orthogonal normals the two projectors coincide, and for $K = d$ non-coplanar boundaries $\mathbf{P}_{\text{tan}} = \mathbf{0}$, so the node is fully enclosed and determined by the boundary data.

A consequence of the renormalisation is that the projection must be performed in physical space, on the raw offset $\mathbf{o}_i$, before renormalisation by $\mathbf{B}_i$. The projected offset is formed first and only then renormalised into the correction vector,

$$\mathbf{o}_{w,i} = \mathbf{P}_{\text{tan}} \mathbf{o}_i, \quad (30)$$

$$\mathbf{V}_i = \mathbf{B}_i \mathbf{o}_{w,i}. \quad (31)$$

The alternative ordering $\mathbf{P}_{\text{tan}}(\mathbf{B}_i \mathbf{o}_i)$ removes a component along $\mathbf{n}$ rather than the correct $\mathbf{B}_i \mathbf{n}$ direction, and the two coincide only if $\mathbf{n}$ is an eigenvector of $\mathbf{B}_i$, which does not occur for realistic clouds. Only the project-then-renormalise ordering of Eq. (31) preserves linear consistency at the boundary, as is shown next.

With $\mathbf{V}_i = \mathbf{B}_i \mathbf{o}_{w,i}$ the correction weight becomes $w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{V}_i$. For a linear field $f = \mathbf{a} \cdot \mathbf{x} + b$, with $f_{ij} = \mathbf{a} \cdot \mathbf{x}_{ij}$, the left-hand side of the assembled row evaluates, using $\mathbf{M}_i \mathbf{B}_i = \mathbf{I}$, to

$$\sum_j W_{ij} f_{ij} w_{ij} = \mathbf{a} \cdot \mathbf{o}_i - \mathbf{a} \cdot \mathbf{M}_i \mathbf{V}_i = \mathbf{a} \cdot \mathbf{o}_i - \mathbf{a} \cdot \mathbf{o}_{w,i} = \mathbf{a} \cdot (\mathbf{I} - \mathbf{P}_{\text{tan}}) \mathbf{o}_i, \quad (32)$$

a residual that is exactly the part of $\mathbf{o}_i$ in the boundary-normal subspace. The closure cancels it with the prescribed flux placed on the right-hand side,

$$b_{N,i} = \begin{cases} g_{\text{eff}} (\mathbf{o}_i \cdot \mathbf{n}_{\text{eff}}), & \text{AN, } g_{\text{eff}} = \sum_k \beta_k g_k, \\ (\mathbf{N}^\top \mathbf{o}_i)^\top \mathbf{G}^{-1} \mathbf{g}, & \text{GGP.} \end{cases} \quad (33)$$

That this is the correct cancellation is verified by writing the Gram projector complement as $\mathbf{I} - \mathbf{P}_{\text{tan}} = \mathbf{N}\mathbf{G}^{-1}\mathbf{N}^\top$ and noting that, for a linear field, the vector of nodal fluxes is $\mathbf{g} = [\mathbf{n}_k \cdot \mathbf{a}]_k = \mathbf{N}^\top \mathbf{a}$. The residual of Eq. (32) is then

$$\mathbf{a} \cdot (\mathbf{I} - \mathbf{P}_{\text{tan}}) \mathbf{o}_i = \mathbf{a}^\top \mathbf{N} \mathbf{G}^{-1} \mathbf{N}^\top \mathbf{o}_i = (\mathbf{N}^\top \mathbf{o}_i)^\top \mathbf{G}^{-1} \mathbf{g} = b_{\text{N},i}, \quad (34)$$

so $\sum_j W_{ij} f_{ij} w_{ij} = b_{\text{N},i} = \nabla^2 f / N_i + b_{\text{N},i}$ holds, with $\nabla^2 f = 0$, on any truncated support, including a non-orthogonal corner. The reversed ordering $\mathbf{P}_{\text{tan}}(\mathbf{B}_i \mathbf{o}_i)$ would replace $\mathbf{M}_i \mathbf{V}_i = \mathbf{o}_{w,i}$ in Eq. (32) by $\mathbf{M}_i \mathbf{P}_{\text{tan}} \mathbf{B}_i \mathbf{o}_i \neq \mathbf{o}_{w,i}$, breaking the cancellation. This closure will be referred to as the *projection closure*.

The projection closure is correct and linear-exact, but it imposes only the first-derivative flux at the boundary. For nonlinear solutions the normal second derivative $\partial^2 p / \partial n^2$ is left to one-sided support and is poorly resolved. As the results of Section 5 show, this makes the Neumann boundary the accuracy-limiter. By the boundary-layer mechanism of LeVeque [54, §2.12], an $O(h)$ boundary-row truncation error sits at a node whose column of the inverse operator is $O(1)$, and is therefore undamped and pollutes the solution with a large constant. The remedy is to complete the boundary stencil, which is the subject of the next section.

3.3. Flux closure (Neumann) by algebraic ghosting

The limitation of the projection closure is removed by stencil completion. Rather than projecting out the unresolved normal direction, it is filled with image samples, such that $\partial^2 p / \partial n^2$ becomes a genuine two-sided difference, with the prescribed flux setting the image values. This is the classical ghost-node, or image-particle, construction [9, 54], and the contribution here is

its packaging for the renormalised operator without additional unknowns. For a Neumann node i with nearby boundary segments $\{(\mathbf{n}_k, \mathbf{x}_{f,k}, g_k)\}$, let R_k denote reflection across the plane through $\mathbf{x}_{f,k}$ with normal $\mathbf{n}_k$, and let the reflection group be the set of non-empty compositions of the R_k . For $K = 1$ the group is $\{R_1\}$, and for an orthogonal corner with $K = 2$ it is $\{R_1, R_2, R_2 \circ R_1\}$, the last composition being the corner double reflection that fills the far quadrant. For a non-orthogonal wedge the group generated by the two planes is in general infinite, and it is truncated to the single reflections and the first-order double reflection $R_2 \circ R_1$. This truncation does not reproduce the exact image set of the wedge, but it is empirically sufficient to restore the symmetry of the moment tensor $\mathbf{M}_i$ and to recover the convergence reported in Section 5.7. Every neighbour j and the node i itself are reflected. A source point $\mathbf{x}_s$ maps under a composition to $\mathbf{x}_s \mapsto \mathbf{x}_s - 2\sigma_k \mathbf{n}_k$ with $\sigma_k = (\mathbf{x}_s - \mathbf{x}_{f,k}) \cdot \mathbf{n}_k$, and the image carries the Neumann-consistent value $p_{\text{ghost}} = p_s + \sum_k (-2\sigma_k g_k)$. Each image within h is appended to the support with its kernel weight and is included in $\mathbf{M}_i$, $\mathbf{o}_i$ and N_i , which restores near-symmetric support. The completed row may be normalised by either form of Section 2. The sum normalisation, Eq. (14), is found to restore close to second-order convergence at a straight Neumann boundary and is preferred, while the smooth-naive form is retained at non-orthogonal corners, as Section 5.3 shows. The Laplacian row then couples the image back to its source node p_s , either a neighbour or i , with the standard weight, and the constant flux increment moves to the right-hand side. Two details are each verified to be necessary. The self-reflection of i supplies the closest and highest-weight image, its diagonal contribution cancelling while its flux increment survives

on the right-hand side. The corner double reflection supplies the otherwise missing quadrant. This closure will be referred to as the *algebraic-ghost closure*. It preserves linear-exactness. For a linear field $f = \mathbf{a} \cdot \mathbf{x} + b$ the flux is $g = \mathbf{a} \cdot \mathbf{n}$, and the assigned image value reproduces the true field at the mirrored point,

$$p_{\text{ghost}} = f(\mathbf{x}_s) - 2\sigma(\mathbf{a} \cdot \mathbf{n}) = \mathbf{a} \cdot (\mathbf{x}_s - 2\sigma\mathbf{n}) + b = f(\mathbf{x}_s - 2\sigma\mathbf{n}) = f(\mathbf{x}_{\text{ghost}}), \quad (35)$$

and the completed support is an ordinary, linear-exact renormalised stencil. Unlike the projection closure, the two-sided support supplies the normal curvature, and the boundary ceases to be the bottleneck. The reflection plane is exact for straight boundary segments. A curved boundary is treated by a finely faceted polygon, with a local-plane generalisation left as future work. The closure is applied only to Neumann nodes. The free surface, the contact lines, and the interior retain the single code path of Sections 3.2 and 3.4.

The ghosts are algebraic and ephemeral. They are formed on the fly within the local neighbour sum of a single row, contribute to that row's moment tensor and coefficients, and are then discarded. No ghost particles are stored, no memory is allocated beyond the row-local arrays, and the global neighbour list is neither extended nor re-searched. This is in contrast to dummy-particle treatments, which allocate and maintain layers of boundary particles and enlarge the neighbour search.

3.4. Value closure (*Dirichlet/Robin*) and free-surface detection

The value closure integrates a Dirichlet or Robin datum into the diagonal of the row, by the principle of Section 3.1. Its showcase is the free surface,

a value boundary that is moreover detected from the cloud rather than prescribed. At the free surface the field satisfies $p = p_{\text{target}}$, which is zero for a clean surface and $\gamma\kappa$ with surface tension. No surface particles are placed, and every node with air-side truncation receives the condition. A Taylor expansion to the surface at a distance δ along the outward normal $\mathbf{n}_s$ converts the Dirichlet datum into a Robin form whose flux depends on the unknown nodal value,

$$\nabla p_i \cdot \mathbf{n}_s = \frac{p_{\text{target}} - p_i}{\delta}, \quad (36)$$

which produces the diagonal modification of Eq. (21) rather than a pure right-hand-side term. The surface geometry is obtained from the cloud as the support deficiency that the Neumann boundary cannot explain, that is, from the projected offset $\mathbf{o}_{w,i}$, such that a corner does not contaminate the surface normal. The residual, the outward normal and the surface distance are

$$\mathbf{r}_i = \mathbf{B}_i \mathbf{o}_{w,i}, \quad (37)$$

$$\mathbf{n}_s = -\mathbf{r}_i/|\mathbf{r}_i|, \quad (38)$$

$$\delta = |\mathbf{o}_{w,i} \cdot \mathbf{n}_s|/S_i. \quad (39)$$

With these natural estimates the Robin enforcement strength $-(\mathbf{o}_{w,i} \cdot \mathbf{n}_s)/\delta$ cancels to a geometric quantity. Writing $\mathbf{n}_s = -\mathbf{r}_i/|\mathbf{r}_i|$ and $\mathbf{r}_i = \mathbf{B}_i \mathbf{o}_{w,i}$, the numerator is

$$\mathbf{o}_{w,i} \cdot \mathbf{n}_s = -\frac{\mathbf{o}_{w,i} \cdot \mathbf{r}_i}{|\mathbf{r}_i|} = -\frac{\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i}}{|\mathbf{r}_i|} < 0, \quad (40)$$

the sign being fixed by the positive-definiteness of $\mathbf{B}_i$, since the offset points inward and the normal outward. The distance estimate is then $\delta = |\mathbf{o}_{w,i} \cdot$

$\mathbf{n}_s|/S_i = (\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/(|\mathbf{r}_i| S_i)$, and the common factor $(\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/|\mathbf{r}_i|$ cancels between numerator and denominator,

$$-\frac{\mathbf{o}_{w,i} \cdot \mathbf{n}_s}{\delta} = \frac{(\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/|\mathbf{r}_i|}{(\mathbf{o}_{w,i} \cdot \mathbf{B}_i \mathbf{o}_{w,i})/(|\mathbf{r}_i| S_i)} = S_i, \quad (41)$$

The enforcement strength is therefore the total kernel weight, which is never singular, is self-scaling, and is free of any penalty parameter. The free surface is detected, again without a threshold, from the dimensionless indicator

$$\lambda_i = |\mathbf{r}_i| \Delta, \quad (42)$$

which is ≈ 0 in the interior and $O(1)$ at the surface, through a smooth activation $\sigma_i = \sigma(\lambda_i) \in [0, 1]$. The cubic smoothstep is recommended,

$$\sigma_i = S(3(\lambda_i - \tfrac{2}{3})), \quad S(t) = \begin{cases} 0, & t \leq 0, \\ 3t^2 - 2t^3, & 0 < t < 1, \\ 1, & t \geq 1, \end{cases} \quad (43)$$

which is zero for $\lambda_i \leq \frac{2}{3}$, leaving the interior nodes untouched, is unity for $\lambda_i \geq 1$, and introduces no parameter. The normal component is then removed in a regularised manner that avoids normalising $\mathbf{r}_i$, and the result is renormalised into the correction vector and weight,

$$\mathbf{o}_{*,i} = \mathbf{o}_{w,i} - \sigma_i \frac{(\mathbf{o}_{w,i} \cdot \mathbf{r}_i) \mathbf{r}_i}{|\mathbf{r}_i|^2 + \eta^2}, \quad (44)$$

$$\mathbf{V}_i = \mathbf{B}_i \mathbf{o}_{*,i}, \quad (45)$$

$$w_{ij} = 1 - \mathbf{x}_{ij} \cdot \mathbf{V}_i. \quad (46)$$

The diagonal then receives $-\sigma_i S_i$ with the right-hand side receiving $-\sigma_i S_i p_{\text{target}}$, where p_{target} is evaluated at the detected surface point $\mathbf{x}_i + \delta \mathbf{n}_s$. The free-surface datum is thereby integrated as the value-dependent Robin flux of

Eq. (36), its known part entering the right-hand side and its value-dependent part the diagonal of the same collocation row, just as the Neumann flux enters the right-hand side. Every boundary condition is in this way recast as a generalised flux and folded into the operator, which is the sense denoted interchangeably by the terms “boundary-integrated” and “flux-integrated” that name the method.

3.5. Contact lines and the unified equation

At a contact line a node is both a Neumann node and a surface node. Because the surface residual is computed from the projected offset $\mathbf{o}_{w,i}$, it lies in the boundary-tangential subspace and the two projections do not interfere. The Neumann projection of Section 3.2 and the surface removal of Section 3.4 compose into a single $\mathbf{o}_{*,i}$ and a single $\mathbf{V}_i$, and Eq. (21) is assembled with the Neumann flux on the right-hand side and the surface term on the diagonal. Interior nodes, with $\sigma_i = 0$ and no boundary, reduce to Eq. (13). Neumann-only nodes reduce to Section 3.2, and surface-only nodes reduce to Section 3.4. The single-code-path property is thereby retained for all node classes. A pure-Neumann domain is singular up to an additive constant and must be closed by a zero-mean constraint rather than a single-point pin. Any free-surface node, with $\sigma_i > 0$, removes the null space automatically.

3.6. A remark on consistency and convergence

The closures inherit the linear consistency of the interior operator. For $f = \mathbf{a} \cdot \mathbf{x} + b$, the gradient (6), the Laplacian (13), the projection closure (33) and the algebraic-ghost closure each reproduce the exact result on any cloud,

including truncated and corner rows, which is verified to round-off in Section 4. The free-surface cancellation (41) is an algebraic identity. As anticipated in Section 2, linear consistency does not imply second-order pointwise consistency under fixed-relative disorder, and the realised order is governed by the boundary closure. The value closure supraconverges, whereas the flux-only Neumann closure, which leaves the normal second derivative one-sided, is the accuracy-limiter by the undamped-boundary-column mechanism of LeVeque [54, §2.12]. The algebraic-ghost closure, by supplying that curvature, removes the bottleneck. These statements are quantified in the following sections. The rigorous consistency and convergence theory of the underlying operators is given in [1, 48].

4. Operator approximation results

The discrete operators were verified against analytic ground truths on regular and scattered clouds. Throughout, the disorder is measured by the chaos level c of [1]. A node is displaced by a random vector with components in $[-c\Delta/2, c\Delta/2]$, and scattered results are reported as the median over at least twenty random realisations. The error measure is the normalised root-mean-square error $\text{Error} = \|\langle f \rangle - f\|_2 / \|f\|_2$ of [1].

The renormalised gradient (6) and Laplacian (13) are linear-exact on every cloud tested, including truncated boundary rows and the non-orthogonal wedge, to $\sim 10^{-15}$ and $\sim 10^{-13}$ respectively. The Gram-projector algebra is exact to $\sim 10^{-16}$. The projector $\mathbf{P}_{\text{tan}}$ is idempotent and symmetric, $\mathbf{P}_{\text{tan}} \mathbf{n}_k = \mathbf{0}$ for every boundary normal, AN and GGP coincide for $K = 1$ and for orthogonal normals, and three non-coplanar boundaries give $\mathbf{P}_{\text{tan}} = \mathbf{0}$.

The Neumann linear consistency $\sum_j W_{ij} w_{ij} f_{ij} = b_{N,i}$ holds to $\sim 10^{-17}$, and the project-then-renormalise ordering of Eq. (31) is required, since reversing it breaks consistency at $O(10^{-3})$ on a jittered cloud. The free-surface cancellation (41) holds to round-off, and a hydrostatic, linear Poisson problem on the wedge tank is recovered to $\sim 10^{-14}$ with GGP boundaries. The algebraic-ghost closure preserves linear-exactness to $\sim 10^{-15}$.

The $2/d$ factor of the smooth-naive normalisation, Eq. (10), is pinned by a three-dimensional constant-Laplacian test, for which the bare $\text{tr } \mathbf{B}_i$ form, without the $2/d$, is in error by a factor of 1.5. The sum operator, with the denominator normalisation N_i^{den} of Eq. (14), being exact for the isotropic quadratic, is between 8 and 15% more accurate at practical resolutions, with its edge widening under anisotropy. The sum-to-smooth-naive error ratio falls monotonically from 1.00 on a lattice to 0.73 at chaos 80%, as shown in Fig. 12. It is offered as a refinement, and the smooth-naive operator remains the default for its lower cost.

The pointwise approximation of the operator was examined on the Franke function of Franke [55] over the unit square, in the manner of [1]. Figure 1 reports the normalised error of the renormalised Laplacian against Δ , on a regular lattice and at chaos 30, 60 and 90%, with two second-order references, the GFDM operator and the same-family Full-Inverse operator of Asai et al. [8]. On a regular lattice all operators are second-order. Under disorder the renormalised operator is linear-consistent, and its pointwise order falls with the chaos level, from 2.0 on the lattice to about 0.7, 0.2 and 0.1 at chaos 30, 60 and 90%, approaching the $O(1)$ truncation error noted in Section 2. The GFDM operator remains close to second-order throughout, and the Full-

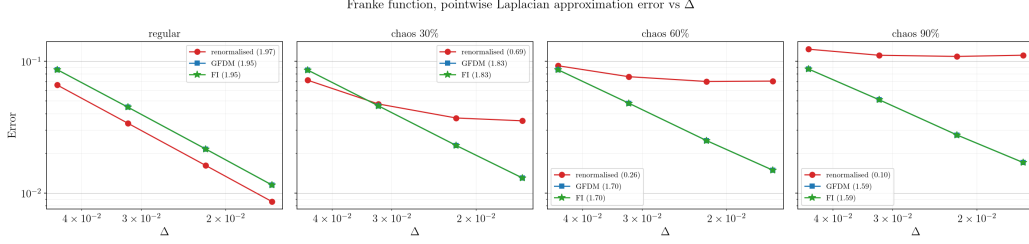


Figure 1: Pointwise Laplacian approximation error of the Franke function against Δ , for the renormalised operator and two second-order references, the GFDM operator and the same-family Full-Inverse operator of [8], on a regular lattice and at chaos 30, 60 and 90%. On a regular lattice all operators are second-order. Under disorder the renormalised operator is linear-consistent and its pointwise order falls with the chaos level, whereas the GFDM and Full-Inverse operators, the latter reinstating the cross-derivative coupling, remain close to second-order and overlie one another.

Inverse operator, which augments the present renormalised stencil with the cross-derivative coupling, tracks it to within plotting accuracy. The two coincide because the lost order is due to the dropped cross terms and not to the renormalised family itself, so recovering it is a matter of reinstating those terms, at the cost of a small local solve per node, rather than of leaving the family. The order of the renormalised operator is recovered by enlarging the support radius. At chaos 60% the order rises to about 1.0 when the support is grown from 2.5Δ to 3.5Δ , and to about 1.8 at 5Δ , at the cost of a wider stencil. The support is nevertheless kept small, because the boundary value problem solution supraconverges at the smaller support, as Section 5 shows.

5. Boundary value problem results

The closures were exercised on a set of Poisson benchmarks on irregular domains, extending the cases of [9, 13] and of [1]. Refinement is by halving Δ ,

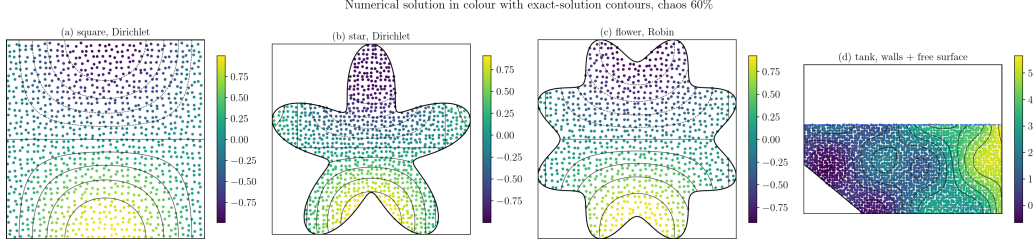


Figure 2: Qualitative solution maps. The numerical solution is carried on the scattered cloud, shown in colour, with the exact-solution contours overlaid as lines, at chaos 60%. From left to right, the square with a Dirichlet condition and the value closure on a flat boundary, the star with a Dirichlet condition and the curved value closure, the flower with a Robin condition, and the wedge tank with GGP Neumann boundaries, the algebraic-ghost closure and a geometry-detected free surface. The contours pass smoothly through the cloud in every case, including at the curved and non-orthogonal boundaries.

and scattered clouds are tested at chaos $c \in \{30, 60, 90\}\%$ with median averaging over at least twenty realisations. The second-order GFDM operator of Section 2.2 serves as the meshless baseline throughout, with its Neumann flux imposed by the exact constraint row and the non-convergent penalty form retained for contrast; for the interior operator comparisons the same-family Full-Inverse operator of Section 2.3 is also included. The non-convergence of the classical SPH Laplacian on irregular clouds was established in [1] and is not repeated. Figure 2 previews the closures on each domain. It shows the numerical solution carried on the scattered cloud against the exact-solution contours, for the square with a Dirichlet condition, the star with a Dirichlet condition, the flower with a Robin condition, and the wedge tank with Neumann boundaries and a detected free surface.

5.1. Square domain with a Dirichlet condition

The unit square with a manufactured trigonometric solution and the Franke test function [55] serves as the interior baseline. Figure 3 compares the renormalised closure with the GFDM baseline and the same-family Full-Inverse operator [8] at the three chaos levels. The two second-order operators retain their pointwise consistency and hold an order near 1.77 across all disorder levels, the Full-Inverse and GFDM curves coinciding, whereas the renormalised operator supraconverges at the linear-consistent ceiling, with an order between 1.4 and 1.5 that degrades only mildly with disorder. The value of the present method does not lie in the interior order, where a fully second-order operator is more accurate under strong disorder, but in the boundary closure examined in the following benchmarks, and in its robustness and absence of tuning; the second-order interior operator and the present boundary closure are complementary and could be combined.

5.2. Neumann boundaries by projection, algebraic ghost and GFDM constraint row

The all-Neumann unit box, closed by a zero-mean constraint, isolates the Neumann treatment. Table 1 and Figure 4 compare four closures, namely the FIRM projection closure, the FIRM algebraic-ghost closure, the GFDM exact constraint row and the GFDM penalty form. The two flux-only closures, FIRM projection and the GFDM constraint row, behave alike. Both impose the flux exactly yet leave the normal second derivative one-sided, and both converge only slowly, at order about 0.9, and carry a large error constant, with the Neumann term dominating the pure-Neumann error. The penalty form is indistinguishable from the constraint row, which confirms that a soft flux is

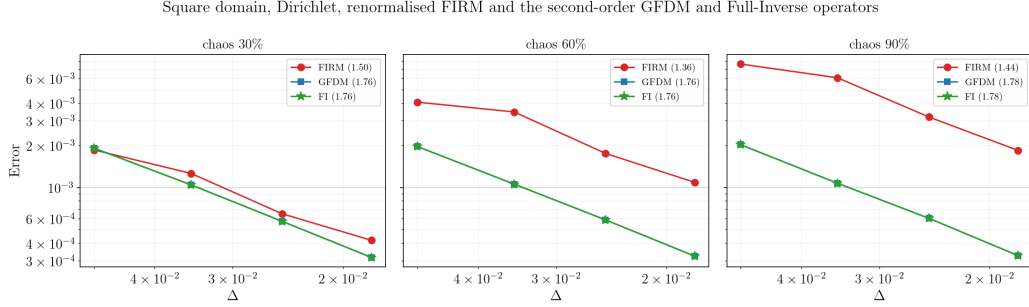


Figure 3: Square domain with a Dirichlet condition and a trigonometric solution. Normalised RMSE against Δ for the renormalised closure FIRM and two second-order baselines, the GFDM operator and the same-family Full-Inverse operator of [8], at chaos 30, 60 and 90%. The renormalised operator supraconverges at an order of about 1.4 to 1.5, while the GFDM and Full-Inverse operators retain their second-order interior order of about 1.8 and coincide.

not the deficiency. The algebraic-ghost closure, which completes the stencil, is about eight times more accurate at the finest resolution, 4.7×10^{-3} against 3.9×10^{-2} . The all-Neumann box is singular up to an additive constant, so its per-closure order is dominated by the null-space handling and is not a reliable rate. The convergence rate is therefore reported on the well-posed wedge tank of Section 5.7, where the ghost closure lifts the Neumann-region order from about 1.0 to between 1.3 and 1.8, so the Neumann region converges at roughly the rate of the interior. It should be noted that the algebraic-ghost closure outperforms the GFDM constraint row here because completing the stencil supplies the curvature that an exactly imposed flux does not. A high-order Neumann treatment would instead require the staggered KKT construction of Trask et al. [29], which is noted as future work.

Table 1: All-Neumann box at chaos 30%, normalised mean-removed RMSE against Δ . The two flux-only closures, FIRM projection and the GFDM constraint row, stall at a large constant set by the one-sided normal curvature, and the penalty form is indistinguishable from the constraint row. The FIRM algebraic-ghost closure is about an order of magnitude lower. The box is singular up to an additive constant, so its per-closure order is set by the null-space handling and is not reported here. The convergence rate is reported on the well-posed wedge tank in Section 5.7. Seed-averaged values from the verification suite.

Δ	FIRM projection	FIRM ghost	GFDM constraint	GFDM penalty
0.050	1.05×10^{-1}	1.1×10^{-2}	1.22×10^{-1}	1.22×10^{-1}
0.035	7.2×10^{-2}	1.6×10^{-2}	8.8×10^{-2}	8.8×10^{-2}
0.025	6.0×10^{-2}	2.2×10^{-3}	6.7×10^{-2}	6.7×10^{-2}
0.018	3.9×10^{-2}	4.7×10^{-3}	4.7×10^{-2}	4.7×10^{-2}

5.3. Normalisation of the ghost closure

The order to which the algebraic-ghost closure converges depends on the normalisation of the completed row. This is isolated on a square with a single straight Neumann edge at $y = 0$ and Dirichlet conditions on the other three edges, which is well-posed and free of the corner truncation of Section 3.3. Figure 5 reports the error on the Neumann edge against Δ . The projection closure converges at order about 0.8. The ghost closure with the smooth-naive normalisation, Eq. (10), improves the constant by an order of magnitude but its error plateaus, at an order of about 1.4. The ghost closure with the sum normalisation, Eq. (14), which is exact for $|\mathbf{x}|^2$, removes the plateau and continues to converge, at an order of about 1.8, approaching second order. On the completed, near-symmetric support the two forms differ only through the residual anisotropy, and the sum form, by reproduc-

Neumann boundaries, mirror-ghost and flux-only closures

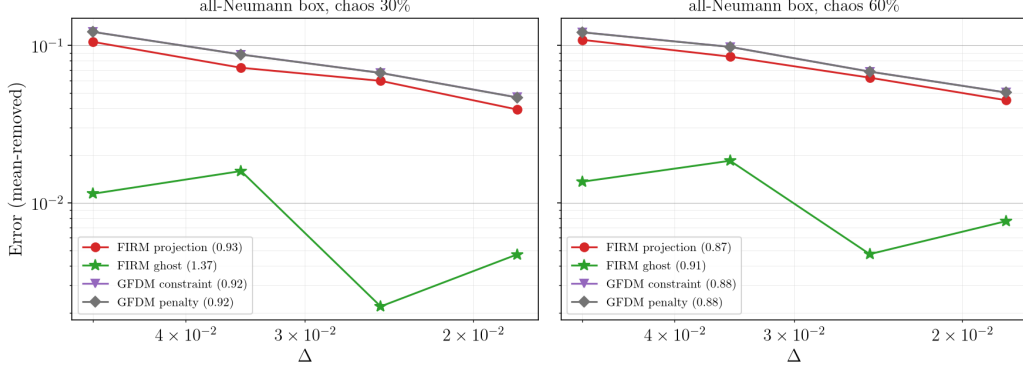


Figure 4: Neumann boundaries on the all-Neumann box. The FIRM algebraic-ghost closure is compared against the flux-only closures, namely FIRM projection, the GFDM constraint row and the GFDM penalty form, at chaos 30% and 60%. Completing the boundary stencil restores accuracy that an exactly imposed flux alone does not.

ing the curvature of $|\mathbf{x}|^2$ exactly, supplies the normal second derivative that the smooth-naive form leaves biased. The sum normalisation is therefore preferred for the ghost closure on straight and smoothly curved Neumann boundaries. At a non-orthogonal corner the rate is instead capped by the truncated reflection group of Section 3.3, where the more robust smooth-naive form is retained, as Section 5.7 shows.

5.4. Star domain with Dirichlet and Neumann conditions

A star-shaped domain $r(\theta) = 0.5 + 0.2 \sin 5\theta$, after [9], with a trigonometric manufactured solution, exercises the closures on a curved boundary. The boundary is finely faceted, and the residual faceting error is held below the solution error. For the Dirichlet condition the value closure of Section 3.4 is used with the local boundary normal and distance, and it supraconverges at order about 1.85, in line with the GFDM baseline at about 1.4. For the

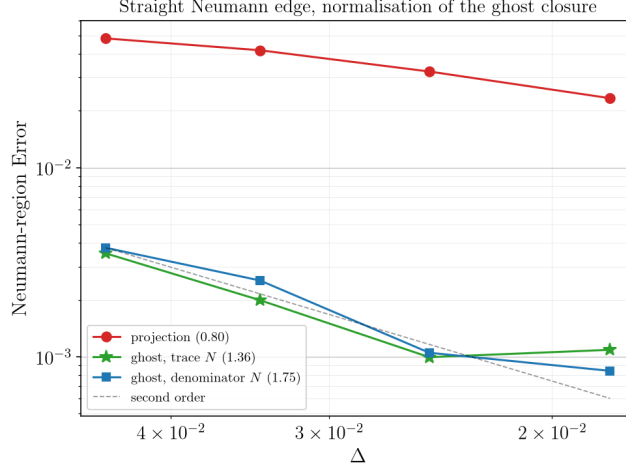


Figure 5: Straight Neumann edge, error on the edge against Δ . The projection closure converges at order about 0.8. The ghost closure with the smooth-naïve normalisation plateaus at order about 1.4, whereas the ghost closure with the sum normalisation approaches second order. Seed-averaged over sixteen realisations.

Neumann condition the projection closure is again the limiter at order about 0.8, while the algebraic-ghost closure improves both the constant and the order to about 1.7. The GFDM constraint row tracks the projection closure at about 0.9, as on the box. Figure 6 collects the curves.

5.5. Flower domain with a Robin condition

A flower-shaped domain with a Robin condition $\partial u / \partial n + \alpha u = f$ exercises the value-dependent flux closure. The boundary value, which is unknown, is eliminated by the local Taylor relation along the perpendicular foot, $u(\mathbf{x}_f) = u_i + \delta \partial u / \partial n$. Substituting this into the Robin condition $\partial u / \partial n + \alpha u(\mathbf{x}_f) = f$ and solving for the normal flux gives

$$g = \frac{\partial u}{\partial n} = \frac{f - \alpha u_i}{1 + \alpha \delta}, \quad (47)$$

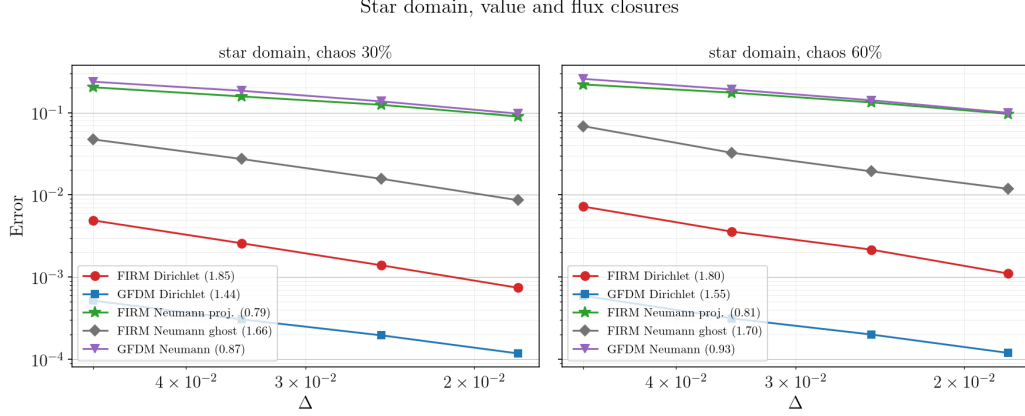


Figure 6: Star domain. The value closure for the Dirichlet condition and the flux closures for the Neumann condition on a curved boundary, at chaos 30% and 60%. The Dirichlet value closure supraconverges, and among the Neumann closures the algebraic ghost is the most accurate.

whose known part $f/(1 + \alpha\delta)$ enters the flux right-hand side exactly as in Eq. (33), while the value-dependent part $-\alpha u_i/(1 + \alpha\delta)$ is linear in the unknown and moves to the diagonal, contributing $\alpha/(1 + \alpha\delta)$ times the geometric weight $(\mathbf{N}^\top \mathbf{o}_i)^\top \mathbf{G}^{-1} \mathbf{1}$. Equation (47) recovers the Neumann closure as $\alpha \rightarrow 0$, where $g \rightarrow f$ and there is no diagonal term, and the Dirichlet value closure as $\alpha \rightarrow \infty$, where $g \rightarrow (f/\alpha - u_i)/\delta$, and it remains linear-exact for straight segments. The realised order rises from about 0.9 at chaos 30% to about 1.1 at chaos 90%, consistent with a flux-type closure, as shown in Figure 7.

5.6. Free-surface detection from the cloud

This benchmark treats the upper boundary of a polygonal tank as an unknown free surface, detected from the cloud by the indicator λ_i and the smoothstep activation of Section 3.4, with no surface particles and no pre-

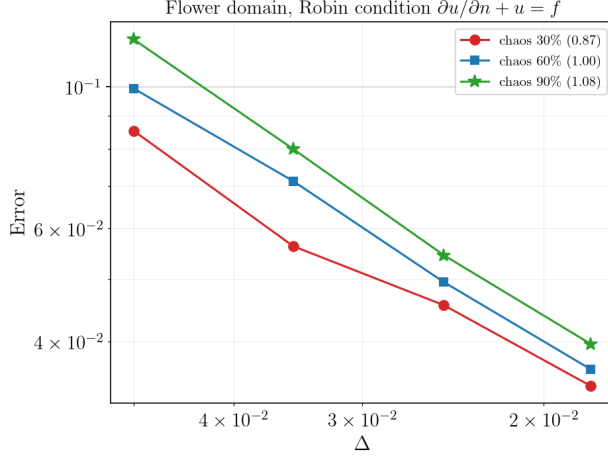


Figure 7: Flower domain with the Robin condition $\partial u / \partial n + u = f$. Normalised RMSE against Δ at chaos 30, 60 and 90%.

scribed geometry. The detector separates the interior from the surface. The interior indicator is $\lambda^{(90)} \approx 0.2$, well below the activation threshold $\frac{2}{3}$. The outermost surface ring has $\lambda \approx 0.7$ to 0.9 . The interior activation is identically zero, with no false activation at any tested resolution or disorder, and the recovered surface normal agrees with the true vertical to within a few percent. The geometry-detected closure tracks a prescribed-boundary reference. The surface-region error of the detected, or natural, closure converges at order about 1.5, close to but slightly below the prescribed, or exact, reference at about 2.2, with no tuning parameter. Figure 8 shows the detected activation field on the tank and the detected-versus-prescribed convergence. For comparison, the rational activation $\sigma = \lambda^2 / (\lambda^2 + c^2)$ with a fixed c over-activates the interior at strong disorder, whereas the smoothstep remains pinned at zero, which motivates the parameter-free choice.

Free-surface detection from the cloud

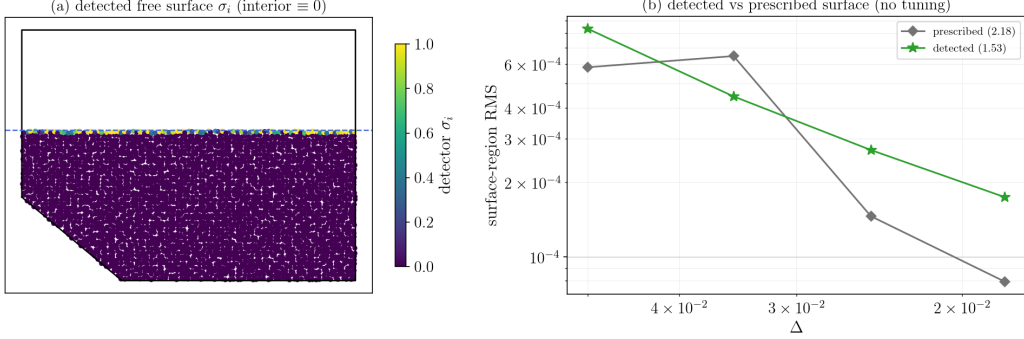


Figure 8: Free-surface detection from the cloud. (a) The geometry-detected activation σ_i on the jittered tank, identically zero in the interior and rising to unity at the surface. (b) The detected, or natural, surface closure converges on top of the prescribed, or exact, reference, with no tuning parameter.

5.7. Non-orthogonal wedge with a free surface

A polygonal tank with a non-orthogonal slant-and-floor wedge and a free surface combines all closures on a complex manufactured field $p^* = \sin \pi x \cos \pi y + 0.3(x^2 + y^2)$. At the wedge the GGP projector is exact for a linear field, with relative error $\sim 10^{-14}$, whereas AN leaks $O(\Delta)$, with relative error $\sim 3 \times 10^{-2}$, as predicted in Section 3.2. For the complex field the error concentrates on the boundaries and the wedge and vanishes towards the free surface, so the Neumann region is the accuracy-limiter under the projection closure, where it converges at order about 1.0. Replacing the projection closure by the algebraic-ghost closure reduces the Neumann-region error by about an order of magnitude, from 1.0×10^{-2} to 7.6×10^{-4} at the finest resolution, and lifts the Neumann-region order to about 1.3 over the full range and about 1.8 over the fine range. The Neumann region thereby converges at roughly the rate of the interior and the Dirichlet boundary, of

Wedge tank, seed-averaged convergence

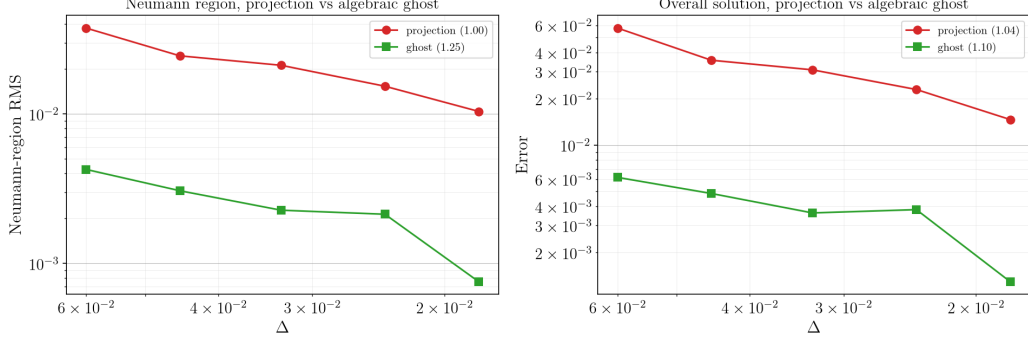


Figure 9: Wedge tank, seed-averaged over six realisations. Left, the Neumann-region RMS for the projection and algebraic-ghost closures. Right, the overall error for the two closures. The ghost closure lowers the Neumann-region error by about an order of magnitude and lifts its order to roughly the interior rate, so the Neumann region is no longer the limiter.

about 1.4 to 1.8, rather than at the reduced rate of the projection closure, so it is no longer the limiter. The errors are seed-averaged over six realisations. The corner double reflection and the self-reflection are both required for this. These results are shown in Figure 9, and the error field in Figure 10.

5.8. Kernel, support radius and disorder robustness

The operators are kernel-agnostic in that linear-exactness holds for the poly6 $(1 - q^2)^3$, the spiky $(1 - q)^3$ and the Wendland-C2 kernels at every support radius. The solution accuracy is not. At matched support the poly6 kernel uses its neighbours most evenly and is the most accurate at small support. For example, at $h = 2.0\Delta$ the Poisson error is 2.3×10^{-3} for poly6 against 6.6×10^{-3} for the spiky kernel. The concentrated spiky kernel is competitive only at $h \gtrsim 2.5\Delta$, where the small-support advantage is lost. The default $h = 2.5\Delta$ is robust, with about fifteen interior and at least

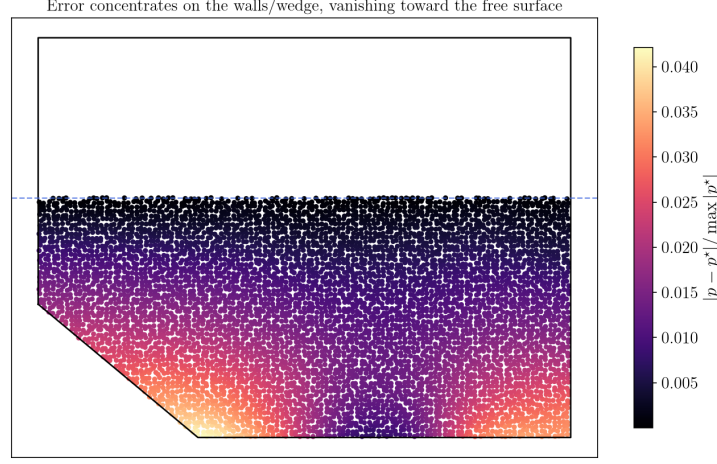


Figure 10: Wedge tank. Pointwise error $|p - p^*|$ for the GGP closure with the prescribed surface at the finest resolution. The error concentrates on the boundaries and the wedge and vanishes toward the free surface, which identifies the Neumann closure as the limiter.

four boundary neighbours at chaos 60%. A smaller $h \in [2.0, 2.25]\Delta$ is viable for quasi-uniform clouds, with the algebraic-ghost closure relieving the boundary-truncation constraint, while $h \leq 2.2\Delta$ hits the two-dimensional rank floor of three neighbours under strong disorder and becomes noisy. Across chaos 0 to 80% the operators remain linear-exact, the condition number of $\mathbf{M}_i$ rises smoothly from 1.9 to about 6 while $\mathbf{M}_i$ stays invertible, the smoothstep activation never false-triggers the interior, and the denominator-normalisation edge widens with anisotropy. These trends are collected in Figures 11 and 12.

6. Application to the Helmholtz–Hodge decomposition

A natural application of the closure is the discrete Helmholtz–Hodge decomposition, in which a vector field is split into a divergence-free part and the

Kernel shape & support radius robustness (chaos 60%)

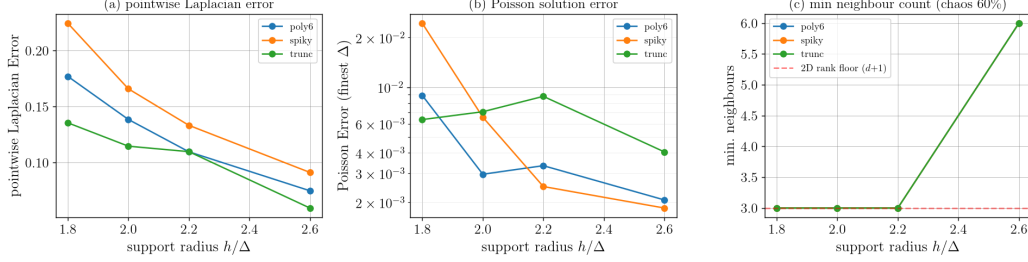


Figure 11: Kernel and support radius at chaos 60%. (a) Pointwise Laplacian error. (b) Poisson solution error, lowest and most robust for poly6 at small support. (c) Minimum neighbour count, showing the two-dimensional rank floor at small radius.

gradient of a scalar potential. The pressure projection of an incompressible-flow solver is its best-known special case, but the decomposition is generic and arises wherever a field must be projected onto its divergence-free part; the treatment here is kept in those general terms. Given a vector field $\mathbf{u}^*$, the decomposition $\mathbf{u}^* = \mathbf{u} + \nabla\phi$ with $\nabla \cdot \mathbf{u} = 0$ is computed by solving a scalar Poisson problem for the potential and subtracting its gradient,

$$\mathbf{L} \phi = \langle \nabla \cdot \mathbf{u}^* \rangle, \quad \mathbf{u} = \mathbf{u}^* - \langle \nabla \phi \rangle, \quad (48)$$

where $\mathbf{L}$ is the renormalised Laplacian with the boundary closure of Section 3, the divergence $\langle \nabla \cdot \mathbf{u}^* \rangle_i = \sum_c \mathbf{e}_c \cdot \langle \nabla u_c^* \rangle_i$ is the trace of the component-wise renormalised gradient (6), and the potential satisfies a Neumann condition at a wall and a value condition at a free surface, by the boundary-condition-by-type principle of Section 3. A concern specific to a meshless, collocated discretisation is that the operators are not mutually adjoint: the single-sum Laplacian (13) is not the composition of the discrete divergence and gradient, $\mathbf{L} \neq \mathbf{D}\mathbf{G}$, where $\mathbf{D}$ and $\mathbf{G}$ denote the divergence and gradient matrices.

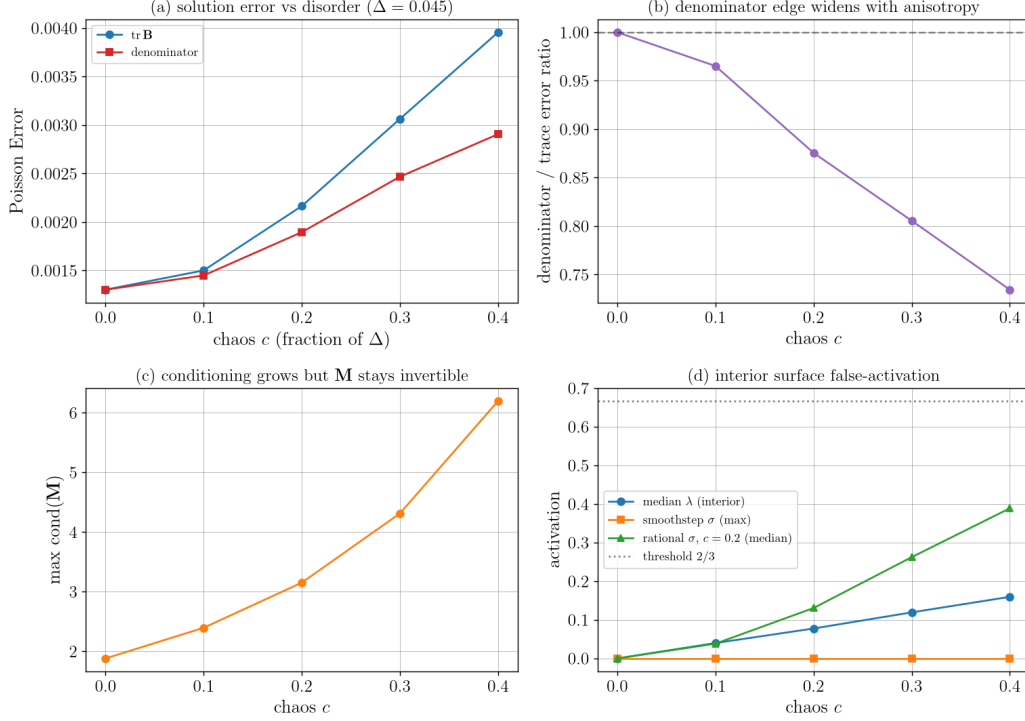


Figure 12: Disorder robustness. (a) Poisson error against disorder for the two normalisations. (b) The denominator-to-trace error ratio narrowing with disorder. (c) The condition number of $\mathbf{M}_i$. (d) Interior surface false-activation, zero for the smoothstep and growing for a fixed-parameter rational activation.

Subtracting the gradient therefore does not annihilate the divergence exactly, and the residual divergence of the computed solenoidal part is, by Eq. (48),

$$\langle \nabla \cdot \mathbf{u} \rangle = \langle \nabla \cdot \mathbf{u}^* \rangle - \text{DG } \phi = \mathbf{L} \phi - \text{DG } \phi = (\mathbf{L} - \text{DG}) \phi. \quad (49)$$

The residual divergence is exactly the operator commutation defect $(\mathbf{L} - \text{DG})$ applied to the computed potential. It vanishes identically for a staggered finite-difference or finite-volume discretisation, for which $\mathbf{L} = \text{DG}$ and the decomposition is exact, and is otherwise a measure of the operators' non-

adjointness. The question is whether this residual is small and whether it vanishes under refinement.

A manufactured Helmholtz–Hodge decomposition provides an analytic ground truth. With a stream function $\psi = \sin \pi x \sin \pi y$ and a potential $\phi^* = \cos 2\pi x \cos 2\pi y$, the field

$$\mathbf{u}^* = \nabla^\perp \psi + \nabla \phi^*, \quad \nabla^\perp \psi = (\partial_y \psi, -\partial_x \psi), \quad (50)$$

has the known divergence $\nabla \cdot \mathbf{u}^* = \nabla^2 \phi^*$, the exact potential ϕ^* up to a constant, and the exact divergence-free part $\nabla^\perp \psi$, which is solenoidal by construction and tangential to the unit-box boundary since ψ vanishes there. The Neumann datum on a wall is the prescribed $\nabla \phi^* \cdot \mathbf{n}$, so the decomposition is exact on any domain, including the wedge tank of Section 5. The discrete divergence $\langle \nabla \cdot \mathbf{u}^* \rangle$ is used as the source in Eq. (48), as it would be in a solver, so that the identity (49) holds discretely.

Equation (49) is confirmed to round-off, the directly measured residual matching $(\mathbf{L} - \mathbf{DG})\phi$ to $\sim 10^{-13}$ on the interior. A single decomposition removes 80 to 95% of the divergence, and the residual converges under refinement, at order 1.2 on a regular lattice, 1.0 on the wedge tank, and 0.4 on the box at chaos 60% with the projection Neumann closure, mirroring the supra-convergent solution orders of Section 5. The algebraic-ghost closure lowers the box residual by about a third and sharply improves the recovered potential, raising its order from 0.9 to 1.7, but it leaves the residual-divergence rate at about 0.4, in contrast to the marked improvement it brings to the Poisson solution rate in Section 5. That a better wall closure improves the potential yet not the divergence rate is a first sign that the residual is limited by the divergence–gradient pairing and not by the Laplacian closure, which the

operator comparison below makes precise. Figure 13(a) shows the residual against Δ for both closures on the box and for the wedge tank. The decomposition is non-expansive, the solenoidal part carrying less energy than the field, $\|\mathbf{u}\|/\|\mathbf{u}^*\| \approx 0.47$, and the discrete inner product $\langle \mathbf{u}, \langle \nabla \phi \rangle \rangle$ between the two parts vanishes under refinement, so the decomposition becomes orthogonal in the limit. The renormalised operators are therefore consistent for a single decomposition: the residual divergence is precisely the non-adjointness defect, and that defect vanishes as $\Delta \rightarrow 0$.

The choice of Laplacian was then isolated. On an all-Dirichlet box, which every operator can solve, the same renormalised gradient was used for $\mathbf{D}$ and $\mathbf{G}$ while the Laplacian $\mathbf{L}$ in Eq. (48) was varied across the uncorrected summation $N_i \sum_j W_{ij} f_{ij}$, obtained by setting $w_{ij} = 1$ in Eq. (13), the present renormalised operator in its smooth-naive and sum normalisations, the GFDM operator of Section 5, and the Full-Inverse operator of Asai et al. [8]. Figure 13(c) reports the residual. The uncorrected summation diverges, its residual growing under refinement, in keeping with the non-convergence of the classical uncorrected SPH Laplacian on irregular clouds [2, 4, 1], so the renormalisation correction is essential to projection consistency. The renormalised operator, the GFDM operator and the Full-Inverse operator are by contrast all consistent and close, the two second-order operators improving the residual only marginally over the linear-consistent renormalised one. The reason is visible in Eq. (49): with a shared gradient the residual $(\mathbf{L} - \mathbf{D}\mathbf{G})\phi$ is set by the gradient pairing $\mathbf{D}\mathbf{G}$ and not by the order of $\mathbf{L}$, so a more accurate Laplacian alone does not reduce it. Projection consistency is thus a property of how the divergence and gradient are paired, more than of the Laplacian's

formal order, which is consistent with the central finding of this work that the closure, and not the interior order, governs the realised accuracy.

The residual can be reduced further by repeating the decomposition: solving Eq. (48) for the current divergence $r_k = \langle \nabla \cdot \mathbf{u}_k \rangle$ and re-correcting, $\mathbf{u}_{k+1} = \mathbf{u}_k - \langle \nabla \delta \phi_k \rangle$ with $\mathbf{L} \delta \phi_k = r_k$. Taking the discrete divergence of the update shows that this defect correction is a fixed-point (Richardson) iteration whose residual propagates by

$$r_{k+1} = (\mathbf{I} - \mathbf{D}\mathbf{G}\mathbf{L}^{-1}) r_k = (\mathbf{L} - \mathbf{D}\mathbf{G}) \mathbf{L}^{-1} r_k, \quad (51)$$

i.e. it is preconditioned Richardson for the exact-projection system $\mathbf{D}\mathbf{G}\phi = r$ with the renormalised Laplacian as preconditioner, and its convergence is governed by the spectral radius of the commutation defect $(\mathbf{L} - \mathbf{D}\mathbf{G}) \mathbf{L}^{-1}$. With the consistent interior source that radius is below but close to unity, so the iteration lowers the residual to a floor of about one to a few percent of the initial divergence and then stalls, never reaching a divergence-free state because the defect has a near-null mode it cannot remove [Fig. 13(b)]. If the raw one-sided divergence at the wall nodes is fed to the source rather than carried by the Neumann condition the radius exceeds unity and the iteration diverges (Fig. 13(b), $\rho_\infty \approx 1.2\text{--}1.5$); this is the flux-only Neumann limitation of Section 3 reappearing in the divergence of the field, and it indicates that the divergence wall term is itself a candidate for the algebraic-ghost completion. The same mechanism limits a fixed-step pressure projection in an unsteady Lagrangian solver, where the source is the rough divergence of a straining, disordered cloud and the unit step amplifies the defect's particle-scale modes.

The fixed unit step is, however, not intrinsic to the decomposition and can be replaced by the step that minimises the residual. Since $\mathbf{D}$ is linear,

correcting $\mathbf{u}_{k+1} = \mathbf{u}_k - \alpha_k \langle \nabla \delta \phi_k \rangle$ leaves the exact divergence $r_{k+1} = r_k - \alpha_k \mathbf{g}_k$ with $\mathbf{g}_k = \text{DG } \delta \phi_k$, so the minimiser

$$\alpha_k = \frac{\langle r_k, \mathbf{g}_k \rangle}{\langle \mathbf{g}_k, \mathbf{g}_k \rangle} \quad (52)$$

gives $\|r_{k+1}\| \leq \|r_k\|$ *unconditionally*, for any cloud and any source, irrespective of the spectral radius in Eq. (51). This minimal-residual (steepest-descent) step is a line search along the same direction the fixed iteration uses and costs one extra divergence evaluation and two inner products per sweep. On the diverging raw-source cases it brings the asymptotic factor from $\rho_\infty \approx 1.2\text{--}1.5$ down to $\rho_\infty \leq 1$, with a residual floor several times smaller than the value the fixed step reaches before it blows up [Fig. 13(b)]; on the consistent interior source it stalls at the same near-null floor as the fixed step, now monotonically. The renormalised closure is therefore suited to a single Helmholtz–Hodge decomposition, with a small residual divergence that vanishes under refinement; iterating it is stable only with the minimal-residual step, which is the form to use when an approximately divergence-free field is sought from disordered or time-evolving clouds.

7. Discussion

7.1. Computational aspects

The closures add only local work to the interior assembly. The Neumann projection is a $d \times d$ projector built from at most K boundary normals, with a single $K \times K$ Gram solve for GGP. The free-surface detection is the matrix-vector product $\mathbf{B}_i \mathbf{o}_{w,i}$, already available from the assembly. The algebraic-ghost closure appends at most a few image samples per Neumann node and

Helmholtz–Hodge decomposition with the renormalised FIRM operators

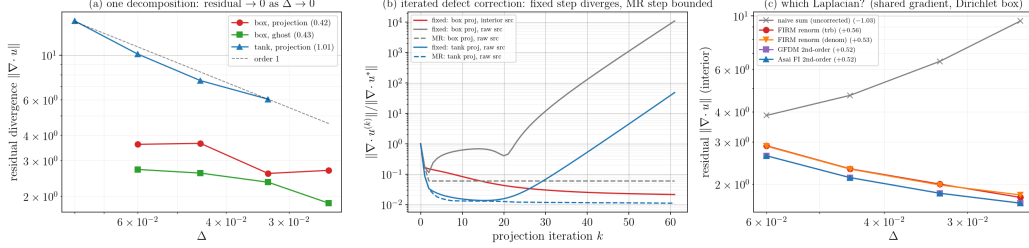


Figure 13: The renormalised operators in a Helmholtz–Hodge decomposition of the manufactured field (50). (a) The residual divergence of the solenoidal part after one decomposition against Δ , on the box with the projection and algebraic-ghost Neumann closures and on the wedge tank, with an order-one reference; it vanishes under refinement. (b) Iterated defect correction at chaos 60%, fixed unit step (solid) versus the minimal-residual step of Eq. (52) (dashed): with the consistent interior source the fixed step falls to a floor and stalls; feeding the raw one-sided wall divergence to the source makes the fixed step diverge ($\rho_\infty > 1$), whereas the minimal-residual step stays non-expansive ($\rho_\infty \leq 1$) and reaches a lower floor. (c) The choice of Laplacian, with a shared gradient on a Dirichlet box: the uncorrected summation diverges, whereas the renormalised, GFDM and Full-Inverse operators are consistent and close, the residual being set by the gradient pairing rather than the Laplacian order.

re-uses the same moment machinery. The boundary work is therefore $O(1)$ per boundary node and is incurred only on the $O(N^{(d-1)/d})$ boundary nodes. The assembled system is sparse and nonsymmetric, because the correction weight w_{ij} and the normalisation N_i are row-local, and a Krylov solver such as BiCGStab or QMR with a Jacobi or ILU(0) preconditioner is appropriate, as in [1]. The recommended configuration is the poly6 kernel at $h = 2.5\Delta$, or $h \in [2.0, 2.25]\Delta$ for quasi-uniform clouds, with GGP projection for corner robustness and the algebraic-ghost closure where Neumann accuracy is required. The choices of AN against GGP and of projection against ghost are

isolated, local code changes, since all downstream assembly consumes $\mathbf{o}_{w,i}$ or the assembled row regardless.

7.2. Operator and closure combinations

Every node is a single row of Eq. (21), so a scheme is just an interior Laplacian together with a closure for each boundary type. Table 2 collects the combinations with their realised convergence orders and per-node cost. The diagonal closures (projection, Robin) require the renormalised operator, whereas the reflection ghosts (even for a Neumann flux, odd for a free surface) pair with any operator and may be mixed with the diagonal forms, so the rows are representative rather than exhaustive. The smooth-naive and sum normalisations of the renormalised operator give the same order with the diagonal closures, sum only lowering the error constant, so the diagonal scheme is listed once; they differ only at the even-ghost Neumann, where the sum normalisation (exact for $|\mathbf{x}|^2$) reaches about 1.8 while smooth naive plateaus at about 1.4, which is why both appear with the ghost closures.

Two of the table’s rows are not, strictly, instances of the unified row, and the exceptions are worth stating. The generalised finite difference operator does not fold the boundary into a single equation: a Dirichlet node is pinned to its value and a Neumann node carries a local Karush–Kuhn–Tucker constraint block, so its system mixes interior, pinned and constrained row types. The Full-Inverse operator keeps one row per node, but that row is the solution of a per-node least-squares system rather than the single renormalised sum of Eq. (21). Both are still free of boundary particles and of extra global unknowns — the reflection ghost is ephemeral and the constraint multiplier is eliminated locally — yet neither is the boundary-integrated single-equation

Table 2: Schemes for the unified row of Eq. (21): an interior Laplacian with a Neumann and a free-surface closure, their realised solution-convergence orders (interior / Neumann / free surface) on disordered clouds, and relative per-node cost. “smooth naive” and “sum” are the trace- and denominator-normalised renormalised operators; “ghost” is the even/odd reflection closure. Orders from Sections 5 and 7.2; cost from Section 7.1. The last column is the fraction of the field’s divergence removed by a single Helmholtz–Hodge projection at the finest resolution tested (Section 6, common renormalised gradient); being set by the divergence–gradient pairing rather than by the Laplacian, it reaches about the same value for every corrected scheme, the uncorrected operators alone failing to clean.

Laplacian	Neumann	Free surface	order (int / Neu / surf)	cost	div. cleaned
smooth naive	projection	Robin	1.4/0.9/1.8	lowest	94%
smooth naive	ghost	ghost	1.4/1.4/2.0	low	94%
sum	ghost	ghost	1.4/1.8/2.0	low	94%
GFDM	constraint	ghost	1.8/0.9/2.0	high	94%
Full Inverse	ghost	ghost	1.8/1.8/2.0	moderate	94%

form, and so neither is FIRM proper: they enter the table as a second-order interior baseline (GFDM) and as the same-family second-order operator (Full Inverse) that the algebraic-ghost closure is shown to transfer to, not as instances of the method. Canonical FIRM is the renormalised operator with the boundary flux integrated into the operator itself — the Neumann flux placed in the right-hand side, and the free-surface value, recast as the value-dependent Robin flux of Section 3.4, split between the right-hand side and the diagonal of one and the same row, with no boundary nodes, no constraints and no auxiliary unknowns. That is the first row of Table 2, the smooth-naive operator with the projection and Robin closures, of which the

ghost rows are the stencil-completing variant, and which the name denotes exactly: a flux-integrated renormalised meshless operator.

Two practical messages follow from Table 2. The renormalised operators introduced here are not the most accurate available: a fully second-order interior, whether the GFDM least-squares operator or the same-family Full-Inverse operator of Asai et al. [8], attains a higher interior order. What the present construction offers instead is convenience. The interior operator is a single neighbour loop with no local linear solve, the boundary conditions are integrated by type on one code path with no auxiliary unknowns and no tuning, and the reflection-ghost closure is operator-agnostic, so a scheme is assembled by choosing one row and one closure per boundary type rather than by deriving a new operator. The cheapest combination in Table 2, the smooth-naive operator with the projection and Robin closures, is also the simplest to implement.

The Helmholtz–Hodge decomposition sharpens this point. The realised cleaning of the divergence is essentially independent of the accuracy ordering: every corrected scheme in Table 2, from the low-order renormalised operator to the second-order GFDM and Full-Inverse operators, removes the same 94% of the field’s divergence in a single decomposition, because the residual is set by the divergence–gradient pairing and not by the interior order (Section 6). The simple, low-order operator is therefore as effective as the higher-accuracy operators for this purpose, at a fraction of the cost, and the small remaining residual is reduced further by iterating the decomposition, which drives the divergence down to a floor of about one percent. That iteration must take the minimal-residual step (Section 6): being preconditioned Richardson on

the non-adjoint divergence–gradient pairing, the fixed unit step can amplify the defect on disordered or time-evolving clouds, whereas the optimal step is non-expansive by construction. A low-order but conveniently assembled operator, combined appropriately and applied iteratively in this way, thus performs the Helmholtz–Hodge decomposition on a par with the higher-order alternatives.

8. Conclusions

A boundary-integrated closure for renormalised meshless operators on elliptic boundary value problems has been presented. First, boundary conditions are integrated by type on a single code path. A Neumann flux enters the right-hand side, a Dirichlet or Robin value enters the diagonal, and the interior, the Neumann boundary, the free surface and the contact lines are assembled by one kernel. Second, the Neumann condition is imposed by a tangential projection performed in physical space before renormalisation, with a general Gram projector that treats arbitrary non-orthogonal corners exactly, and a complementary algebraic-ghost closure that completes the truncated boundary stencil without additional unknowns and removes the Neumann accuracy bottleneck. Third, the free surface is detected from the support-deficiency geometry of the cloud and enforced without any tuning or threshold parameter, the enforcement strength cancelling to the total kernel weight.

The closures are linear-exact on any cloud, including truncated and corner rows, as verified to round-off, and they were tested on irregular-domain Poisson benchmarks for regular and highly irregular clouds against a second-

order GFDM baseline. The convergence is reported in full. The renormalised interior operator is linear-consistent and supraconverges at order about 1.3 under strong disorder, where a fully second-order GFDM operator is more accurate. The value closure supraconverges, the flux-only Neumann closure is the accuracy-limiter, and the algebraic-ghost closure reduces the Neumann-limited error by about an order of magnitude and lets the Neumann region converge at the interior rate, approaching second order at a straight boundary with the sum normalisation. The value of the method lies in the generality, robustness and absence of tuning of its boundary closure, and in the convenience of its assembly, rather than in a high formal order. The interior operator is a single neighbour loop without a local solve, and the closures are selected by boundary type on one code path, so a scheme is assembled by combination rather than by deriving a new operator (Table 2).

Several applications and extensions are topics for future work. The boundary closure is generic for elliptic boundary value problems on moving point clouds. A leading application is the Helmholtz–Hodge decomposition, examined in Section 6, in which the Neumann condition arises at solid boundaries and the free surface carries the value condition, with the pressure projection of an incompressible flow as one special case. There the operators are shown to be consistent for a single decomposition, with the residual divergence equal to their non-adjointness defect and vanishing under refinement, and governed by the divergence–gradient pairing rather than the Laplacian order. In consequence the simple low-order operator cleans the same fraction of the divergence in a single decomposition as the second-order GFDM and Full-Inverse operators, and an iterated decomposition with the minimal-residual step

— which keeps the iteration stable on the disordered, time-evolving clouds where a fixed step would diverge — reduces the residual further, so the conveniently assembled low-order construction performs the Helmholtz–Hodge decomposition on a par with the higher-accuracy alternatives (Table 2). A full incompressible-flow application would additionally require the no-slip velocity Robin condition and the stress-free surface, which inherit the same boundary-condition-by-type treatment, and a divergence-free time integration; these were not assessed here. The motivating application is incompressible free-surface hydrodynamics — sloshing, dam break, and wave impact on moving point clouds — where the cloud-detected free surface, requiring neither surface particles nor a threshold, is the principal advantage. A distinct near-term application is the processing of scattered experimental velocity fields: pressure reconstruction from particle image velocimetry, in which a pressure Poisson problem is solved on a masked, mesh-unfriendly domain with Neumann data from the momentum balance at solid boundaries and a value condition at a free surface or far field, and the Helmholtz–Hodge filtering of noisy measured fields onto their divergence-free part. Fluid–structure interaction and other moving-boundary problems are further candidates for the same Lagrangian, geometry-detected treatment. A locally second-order-consistent interior operator, such as the Full-Inverse renormalised operator of Asai et al. [8] that reinstates the cross-derivative coupling, lifts the whole-domain order. Paired with the reflection-ghost closure — even reflection for a Neumann wall, odd reflection for a Dirichlet free surface, the two composed at a contact line — which transfers unchanged because it is operator-agnostic stencil completion, it yields about second order throughout: a second-order

interior, a Neumann boundary at order about 1.8, a detected curved free surface at order about 2.0, and contact lines at the interior rate, with no extra unknowns (Section 7.2). The same reflection ghosts apply to the linear-consistent renormalised operator, where the odd-ghost free surface in fact surpasses its diagonal-Robin value closure. A high-order flux-only Neumann treatment through the staggered KKT construction of Trask et al. [29] or the GFDM constraint row [27, 28] remains for future work. A curvature-aware reflection (the local planes are flat facets here), surface tension through a meshless curvature estimate, and three-dimensional validation are also left for future study.

Acknowledgements

Placeholder.

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